

SyncRun – Align Music to Run

Matan Granot, Yogev Grinblatt

Advisor: Dr. Lior Arbel

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Introduction

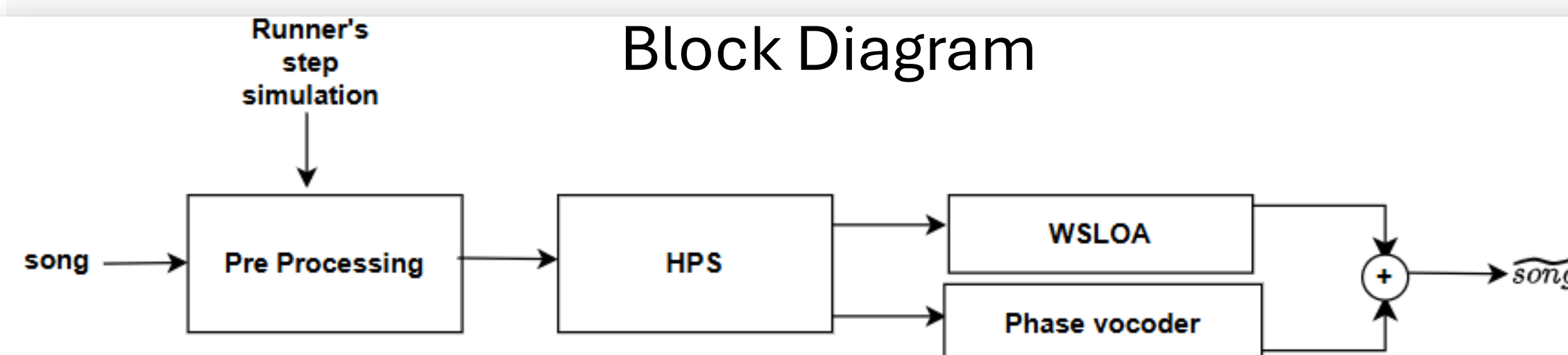
- In our project, we want to adjust a simple rhythmic song – like “Seven Nation Army” to match the simulated steps of a runner, during his various workouts.
- This is done by using DSP algorithms, to find the optimal speed-up/speed-down rate, and to stretch the audio by the best value, while keeping the pitch of the original song. These algorithms are implemented in python.

Motivation

- We want to give the runner smoother running experience. Because in reality, most of the songs are slower than the average runner pace.

Implementation

Block Diagram

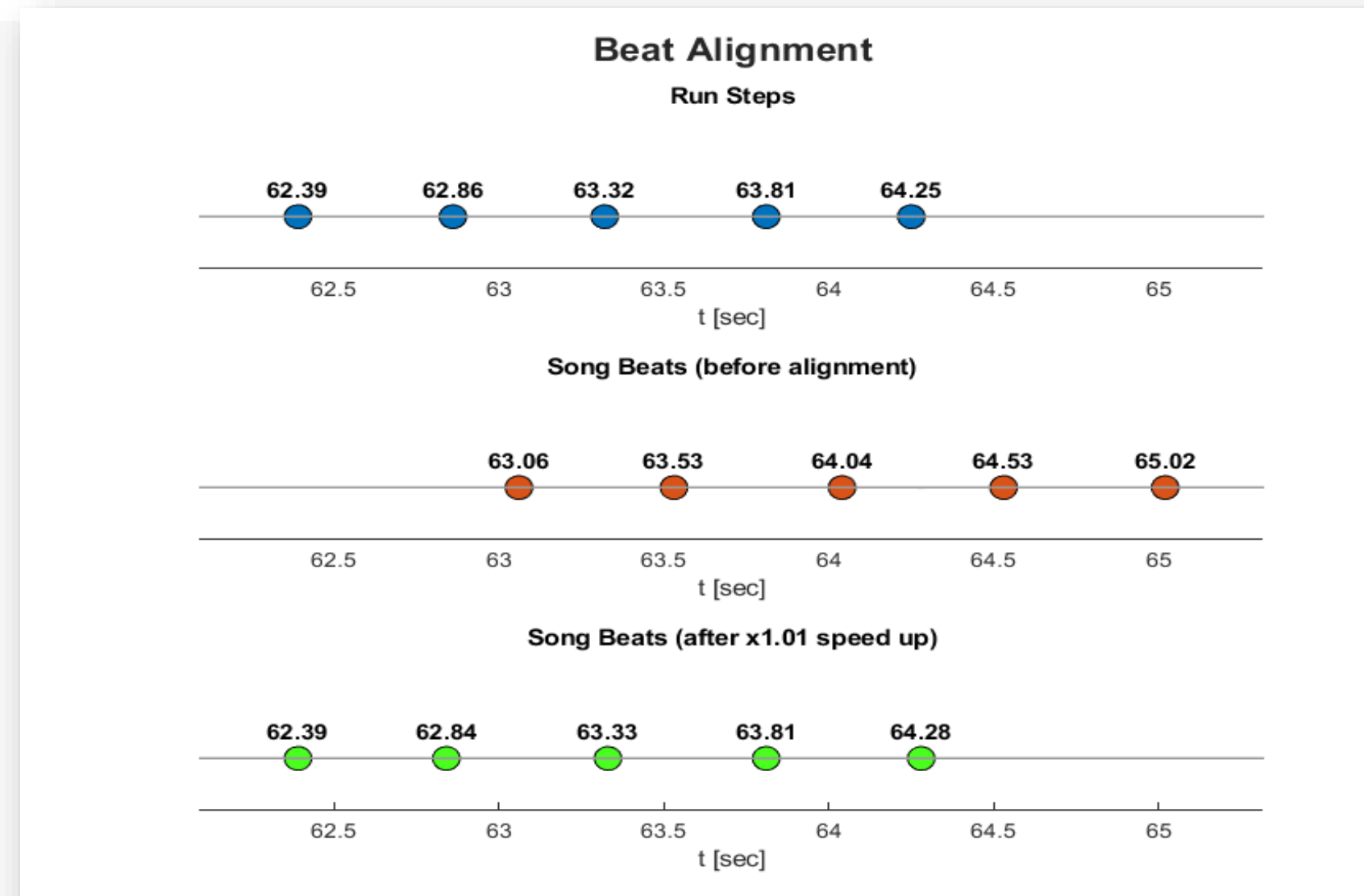


Pre-processing:

First, we make 2 time-stamped vectors – One song beats time stamps (using beat track), Second is run steps time stamps.

Then we do beat alignment - divide those vectors in segments of 2.5 seconds, and find the best speed-up ratio for each segment, in terms of L1 distance.

$$r_{opt} = \underset{r}{\operatorname{argmin}} \{ \|\overrightarrow{beats} - r_{speed-up} * \overrightarrow{steps}\| \}$$

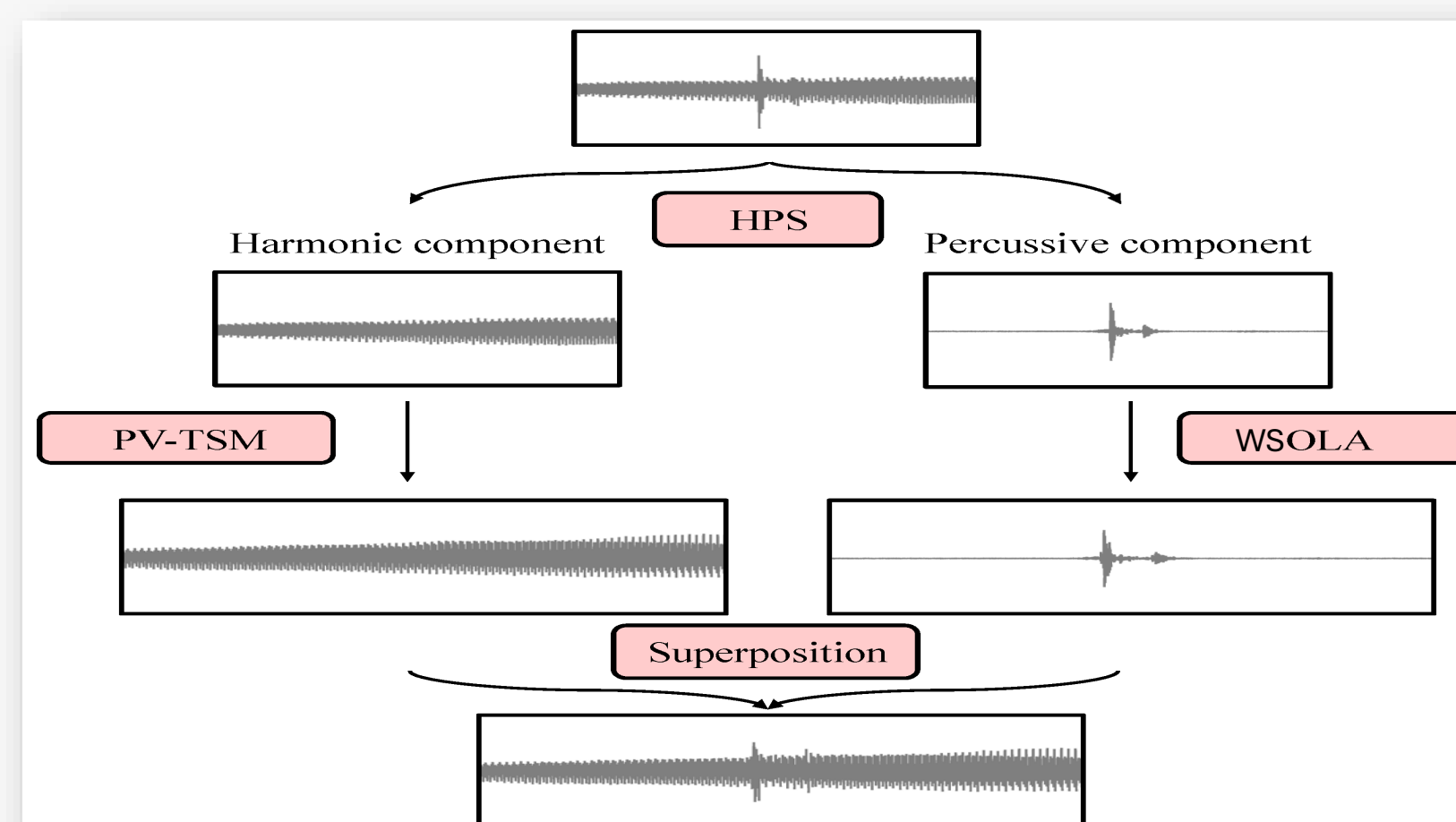


- HPS (Harmonic-Percussive Separation):** we apply STFT (Short Time Fourier Transform) :

$$X[m, k] = \sum_{n=0}^{N-1} x[n + mH] \cdot w[n] \cdot e^{-j\frac{2\pi}{N}kn}$$

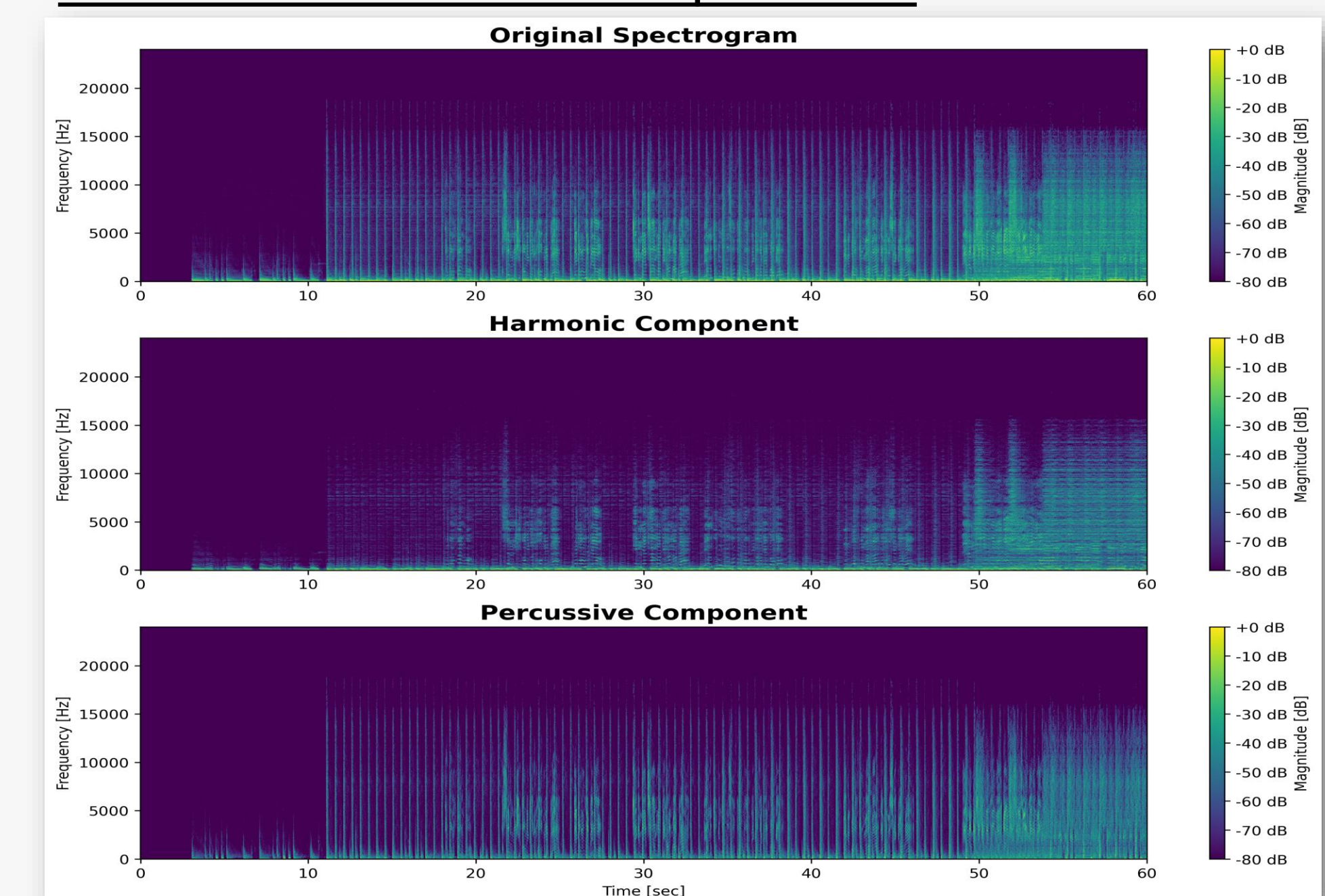
And use median filters to separate the harmonic and percussive content.

- WSOLA** - Changes audio duration without altering pitch by overlapping similar time-domain waveforms.
- Phase vocoder (PV-TSM)** - Independently controls time and pitch using the STFT to adjust frequency phases.



Results

- Harmonic – Percussive Separation:**



Our algorithm successfully splits the original audio into its sustained, pitched sounds (horizontal lines in the Harmonic plot) and sharp, transient drum hits (vertical lines in the Percussive plot).

- Performance:**

Criteria	Result
Average Beat Alignment Error	15.67 ms
Average Segment Latency	475 ms
Elapsed Time	38.47 sec

References

- [1] Jonathan Driedger and Meinard Müller, “A Review of Time-Scale Modification of Music Signals”, Applied Sciences (2016), volume 6, article 57.